Vehicle Trace Detection in Two-Pass SAR Coherent Change Detection Images With Spatial Feature Enhanced Unet and Adaptive Augmentation

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***Abstract*— As a typical application of remote sensing technology, change detection can find the ground information changes by acquiring the images of the same region at different times. The change detection using the synthetic aperture radar (SAR) with the advantages of all day and all-weather usually monitors the significant surface change, such as flood disasters and earthquake deformation. However, when it comes to detecting subtle changes such as vehicle traces, the traditional methods ignoring the phase coherence between image pairs cannot intensify these faint changes in the difference image. The SAR coherent change detection (CCD) based on repeat-pass repeat-geometry complex images utilizing both the intensity and phase fraction could exhibit the subtle vehicle trace in the difference image. However, the complicated background and decorrelation factors significantly affect the quality of difference images, further causing great trouble for automatic trace detection. This article proposes the spatial feature enhanced Unet and adaptive data augmentation to realize vehicle trace detection. More specifically, the pseudocolor image is first synthesized based on a two-stage coherence estimation method. Then, considering the long-continuity and parallel distribution of vehicle trace samples, the enhanced Unet is constructed by fusing spatial convolutional neural network and spatial attention mechanism. After that, the adaptation data augmentation strategy is presented by introducing manual registration errors and multiple estimation windows. Finally, the experimental results on the Sandia CCD data and our measured data demonstrate the effectiveness of the proposed method.**

***Index Terms*— Attention mechanism, coherent change detection (CCD), convolutional neural network (CNN), data augmentation, synthetic aperture radar (SAR).**

I. Introduction

With the advantages of all-weather and all-day, synthetic aperture radar (SAR) can obtain high-resolution remote sensing images from long operation distances by transmitting large-bandwidth signals and growing synthetic arrays [1], [2]. The complex-valued SAR images can be attained using the canonical imaging algorithm to process the complex-valued echo data [3], [4]. The intensity fraction represents the reflection energy of various landmarks, and the phase fraction shows the electromagnetic characteristics, such as geometric structure and frequency response [5]. The intensity fraction of SAR images has been widely used in image information interpretation, such as landmark classification [6], [7], target discrimination [8]–[10], and change detection [11]. In this article, we use both the intensity and phase formation of SAR images to detect emerging subtle vehicle traces between repeat-pass SAR images, named coherent change detection (CCD) [12], [13].

According to the difference of utilized SAR image features, the change detection in SAR images can be categorized into incoherent change detection (ICCD) and CCD methods. The ICCD methods try to extract the intensity difference between SAR images, which are usually used to monitor large-scale landmark variants, such as flood disasters and earthquake deformation [14], [15]. Different from ICCD, the SAR CCD methods [16]–[18] based on repeat-pass repeat-geometry imaging conditions utilize the complex coherence of magnitude and phase information within a local window between SAR image pairs. Related to the system's working frequency and devised imaging resolution, the commonly used millimeter-wave SAR system has the potential to detect very subtle scene changes to the subresolution cell, such as footprints and vehicle traces [16], [19]. In this article, we combine CCD method and automatic detection strategy to extract subtle vehicle traces.

The traditional SAR CCD processing methods usually consist of two steps: the subpixel registration of repeat-pass repeat-geometry SAR image pairs and the accurate complex coherence coefficient estimation based on the well-registered images [12], [18]. For the first step, the baseline decorrelation, or the residual uncompensated phase errors in primary and repeat-pass images, will lead to undesirable image decorrelation and co-registration errors [20]. Since the CCD methods could detect subtle scene changes to the coherent subresolution cell scattering structure, the registration accuracy highly affects the final detection accuracy [16], [21]. After attaining the registered image pairs with subresolution accuracy, the coherence estimation generates the difference images between

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image pairs. ICCD methods usually use the mean backscatter power difference or the intensity ratio to create a difference image [14], [22]. Unlike ICCD methods, the CCD methods must consider SAR images' phase fraction and intensity fraction. Generally, the complex-valued coherence measurement method based on the maximum likelihood can accurately estimate the coherence between SAR images [12], [17], [23]. This method is applied under an assumption of complex Gaussian data of SAR images. The generated difference image with a value between 0 and 1 reflects the complex-valued correlation. The higher the value, the higher the correlation and vice versa. For the vehicle trace concerned in this article, the vehicle moving leads to subtle surface deformation, further causing a low coefficient value in the difference images [13], [16], [20]. Thus, we try to use the low coherence of vehicle traces to realize automatic detection.

Although the maximum-likelihood method can detect the subtle surface change such as landfill activity and vehicle traces, the complex background clutter, such as water area change and vegetation swing, could lead to a low coherence coefficient in the difference image, further causing false alarm in the final vehicle trace detection result [16], [17]. In [23], the statistics of the Stokes parameters and the phase difference are derived as a function of the mean phase difference and the degree of coherence SAR images. Assuming that the variances of the populations underlying each image in the pair to be equal, Cha *et al.* [24] investigated the suitability of the Berger coherence estimator for SAR CCD. In [17], the complex reflectance change detection (CRCD) metric is used for temporal change detection considering the low clutter-to-noise power ratio within the image pairs. Especially, Cha *et al.* [19] proposed the two-stage detection method, which combines the intensity ratio statistics and the Berger estimator. The combined test provides better detection results than either statistic alone. Although the above techniques can suppress the false alarm samples and highlight the interesting trace region in the difference image [20], [25], the vehicle trace cannot be extracted automatically from the cluttered background. Some papers try to introduce more image information into the coherent estimation, such as the multipass [26] and multiband [27] CCD method. Although these methods can effectively distinguish artificial traces from false alarm targets, multiple flights or multiband lead to a complicated operation burden. Moreover, some natural areas with strong time decorrelation still show false alarms in the detection results.

After attaining the difference images between paired images, the main problem is how to extract interesting change regions accurately and automatically from the cluttered background [19], [28]. For the ICCD methods, which try to detect large-scale landmark changes, such as earthquake deformation and flood disaster, the difference image analyzing processes can be divided into traditional nondeep learning and deep learning methods [22], [29]. The conventional techniques usually use the contour features, intensity distribution, and boundary structure of difference images to recognize changed or unchanged pixels [30], while the deep learning-based methods try to extract semantic and abstract feature from the difference image using various network structures. The restricted Boltzmann machine [8] was proposed to learn the feature relationships between paired images for the deep learning-based change detection methods. This network is unrolled to classify each pixel in the difference image. Liu *et al.* [31] presented a symmetric convolutional coupling network that consists of several coupling layers and convolutional layers. In [32], the recurrent neural network and convolutional neural network (CNN) are combined into one end-to-end network, which can learn a joint spectral–spatial–temporal feature representation. Liu *et al.* [33] improved the original CNN with local spatial constraints and proposed the local restricted framework for change detection. These change detection methods can effectively utilize the change information of large-scale landmark information and accurately extract interesting areas from the difference image. Unlike ICCD methods, the main challenge of CCD is how to extract the subtle trace changes like vehicle trace from the blurred difference image caused by co-registration error and complex incoherent clutter. Accordingly, if these methods are directly introduced into the automatic vehicle detection of CCD images, the natural trace features such as continuity and parallelism are ignored [34]. Thus, they may fail to complete disconnected vehicle traces caused by radar noise and various environmental effects, such as vegetation and weather. To overcome these problems, Chow and Quach [34] proposed a simple convolutional network that consists of a series of 3×3 convolutions and dilated convolutions, which can increase the receptive field of the network without sacrificing resolution. Zhang *et al.* [35] proposed a lightweight compressed Unet and corresponding fitting strategies to realize automatic trace detection from CCD images. Although these methods consider the natural properties of the trace difference image, the utilization efficiency of trace features can still be improved.

In this article, considering the natural properties of vehicle trace in CCD images, a proposed detection method combines the spatial feature enhancement Unet and adaptive augmentation. The main contributions are shown as follows.

1) The three-channel pseudocolor image is constructed based on original repeat-pass SAR images and the difference image generated by coherence estimation. Compared with the similar CUnet method, the proposed pseudocolor image avoids complex false alarm elimination processing; meanwhile, the difference information, including vehicle trace and the landmark information, is retained, which can simplify the detection procedure and improve the detection performance.

2) A novel detection network based on fully convolutional Unet is proposed focusing on the long-range connectivity and parallel distribution of vehicle trace. Compared with the original Unet structure, the spatial CNN (SCNN) in our detection network captures vehicle trace's long-range properties and predicts vehicle results with better connectivity and more explicit boundary information. Meanwhile, considering the sparse distribution of the vehicle trace at the large-scale background region, the spatial attention mechanism is introduced into our detection network to pay more attention to the vehicle trace.

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3) A novel CCD image augmentation strategy is devised to train a good vehicle detection network. For preparing the optical image-based detection network, flipping and rotation are commonly used to augment the detection dataset and improve the corresponding performance. Unlike visual images, the vehicle trace generated by CCD processing involves repeat-pass image co-registration and coherent estimation. To improve the performance under a complicated background, the multiple registration errors and various coherence estimation windows are adaptively introduced into our method to augment vehicle training samples of vehicle trace.

The remainder of this article is organized as follows. Section II introduces the related works of the proposed method. Section III describes the proposed framework for trace detection in detail. Section IV shows the experimental results on Sandia CCD data and our measured SAR data. Section V concludes this article.

II. Related Work

The proposed methods are related to the statistical model of SAR CCD and the fully CNN. Some works that inspired our work are shown as follows.

A. SAR Coherent Change Detection Statistical Model

Assuming f and g are two registered SAR images with subpixel registration accuracy of N pixels, they can be constructed as a joint vector $X = [f, g]^T \in \mathbb{C}^{N\times 2}$. The pixels in these images are usually viewed as independent samples from zero-mean bivariate complex Gaussian distribution [12], [17], [23]. The covariance matrix usually follows:

$$\Sigma = \mathbf{E}\left(XX^H\right) = \begin{bmatrix} \sigma_f^2 & \rho\sigma_f\sigma_g \\ \bar{\rho}\sigma_f\sigma_g & \sigma_g^2 \end{bmatrix} \tag{1}$$

where $\sigma_f^2 = \mathbf{E}(|f|^2)$, $\sigma_g^2 = \mathbf{E}(|g|^2)$. Here, ρ is the complex coherent coefficient, and $\bar{\rho}$ is the complex conjugate of ρ. The maximum-likelihood estimator is used to estimate the covariance matrix $\sum$

$$\sum_{k=1}^{N} X_k X_k^H = \begin{bmatrix} \sum\limits_k \left|f_k\right|^2 & \sum\limits_k f_k g_k^H \\ \sum\limits_k f_k^H g_k & \sum\limits_k \left|g_k\right|^2 \end{bmatrix} \triangleq A \tag{2}$$

whose distribution is given by a bivariate complex Wishart density [19]

$$p\left(A; \sum, N\right) = \frac{\left|A\right|^{N-2} \exp\left(-tr\left(\sum^{-1} A\right)\right)}{\pi\,\Gamma(N)\Gamma(N-1)\left|\sum\right|^N} \tag{3}$$

where A is usually obtained using a spatial window of f_k and g_k in the spatially registered images. There are several complex coherence estimators, and some of them are

$$\hat{\rho}_c = \frac{A_{12}}{\sqrt{A_{11}}\sqrt{A_{22}}} \tag{4}$$

$$\hat{\rho}_a = \frac{2A_{12}}{(A_{11} + A_{22})} \tag{5}$$

where $\hat{\rho}_c$ is the most classical estimator, whose distribution only depends on the true underlying coherence magnitude and the number of samples in the estimation window [12]. $\hat{\rho}_a$ is the Berger coherence estimator that assumes variance equality between paired images to improve detection performance [19]. Besides these estimators, the CRCD metric is also presented [17]

$$\hat{\rho}_{\mathrm{CRCD}} = \frac{2A_{12}}{\left(A_{11} + A_{22} - N\delta_f^2 - N\delta_g^2\right)} \tag{6}$$

where δ_f^2 and δ_g^2 are the clutter power (signal power reflected by the clutter of the scene in the local window with N samples) of f and g, respectively.

The above estimators have different properties. $\hat{\rho}_c$ has well robustness, which can detect more minor changes under different imaging scenes. For scenes where most of the underlying variance remains unchanged, $\hat{\rho}_a$ can improve the performance over $\hat{\rho}_c$. The $\hat{\rho}_{\mathrm{CRCD}}$ considering the changing clutter-to-noise ratio (CNR) levels and improve the differentiation between changed and unchanged pixels. Thus, the choice of coherence estimator significantly affects vehicle detection. This article uses a two-step change detection to capture the intensity change and complex coherence change between two-pass images [19].

B. Fully Convolutional Networks for Image Classification

As a typical network structure, the CNN has achieved notable success in the field of natural image processing, such as target classification [36], image detection [37], and semantic segmentation [38]. The fully connected layers in CNN cover lots of weighting parameters and lead to huge complexity. To overcome these problems, the global pooling layers and 1×1 convolution layers are introduced to replace fully connected layers, and the generated network is named the fully convolution networks (FCNs) [39]. FCNs have been widely used in semantic segmentation, where each pixel in one image is classified as the corresponding target category or the background class. FCNs-based Unet [38], [40] is also extended into the pixelwise classification of remote sensing images with flexible structure and fast inference speed. Combined with the encoder module and decoder module, the Unet hierarchically connects feature maps in different convolutional layers with the skip connection.

Considering the difference between optical images and SAR images, some papers try to modify the FCNs and Unet to improve the classification performance. The main changes are the training dataset, network structure, loss function, and learning strategy [32], [41], [42]. These methods could attain superior results of SAR landmark classification and change detection. However, when it comes to vehicle trace detection in the CCD images, the properties of these traces, such as long-continuity and parallel distribution, should be noticed. Although Zhang *et al.* [35] proposed a lightweight simplified Unet to extract trace samples from CCD images, this network devised for the few-shot learning still ignores the unique characteristics of trace samples. In this article, we collect more trace samples, and the main challenge is how to use these samples to attain a detection model with higher detection performance.

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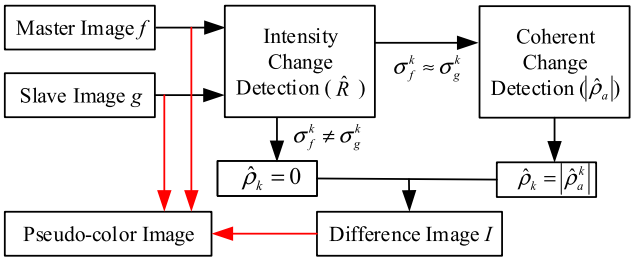


Fig. 1. Generation of pseudocolor image.

III. Proposed Method

The proposed method consists of three key modules: 1) pseudocolor image synthesizing based on a two-stage detection method; 2) Unet-based vehicle trace detection network with SCNN and spatial attention mechanism; and 3) adaptive augmentation of trace samples based on multiple registration errors and estimation windows.

A. Pseudocolor Image Synthesizing

The acquired two-pass repeat-geometry SAR images, which are named master image and slave image, respectively, should be preprocessed to generate a difference image. To be specific, the frequency interpolation method is first used to adjust the azimuth resolution of the slave image to suppress the flight velocity difference between two flights. Then, the real correlation coefficient registration method realizes the coarse registering of these images at up 1- or 2-pixel accuracy [12]. Then, the cutting-edge optical flow method realizes fine registering with subpixel accuracy [43], [44]. Finally, resampling the slave image according to subpixel transformation, we can get the registered master image f and slave image g. These images can be used for subsequent coherence estimation and trace detection.

The generation of pseudocolor image based on change detection is shown in Fig. 1. We first use the noncoherent intensity change detection to detect the pixels with large-scale change. To be specific, the clutter power at one location k in master image f and slave image g is estimated within a local window, respectively, and the intensity estimator is

$$\hat{R}_k = \hat{\sigma}_{f,k}^2 / \hat{\sigma}_{g,k}^2. \tag{7}$$

The probability distribution function (pdf) is usually modeled as [19]

$$p\left(\hat{R}; R, |\rho|, N\right) = \frac{\hat{R}^{N-1}\left(\hat{R}+R\right)R^N\left(1-|\rho|^2\right)^N}{B(N,N)\left[\left(\hat{R}+R\right)^2 - 4R\hat{R}|\rho|^2\right]^{N+1/2}}, \quad \hat{R} \geq 0 \tag{8}$$

where $B(N, N)$ is the beta function. R is the true variance ratio of σ_f^2 and σ_g^2. If the value $\hat{R}$ of one pixel k differs substantially from the others, this pixel is labeled as change. More formally, given a boundary level $\alpha \in (0, 1)$, if the sample value $\hat{R}_k < R_{l,\alpha}$ or $\hat{R}_k > R_{u,\alpha}$, where $R_{l,\alpha}$ is the low boundary and $R_{u,\alpha}$ is upper boundary decided by α, this sample indicates significant scale change, such as the position change of one car [19].

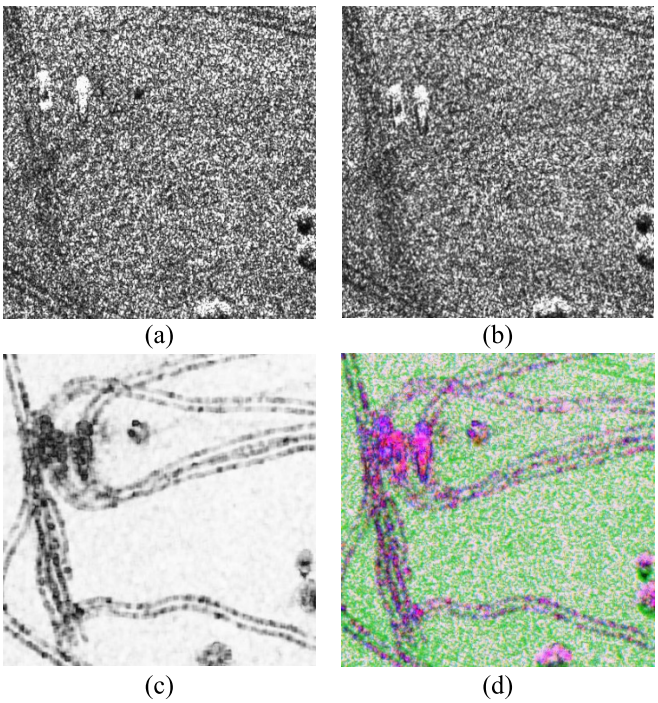


Fig. 2. Constructing of pseudocolor image. (a) Master image. (b) Slave image. (c) Difference image. (d) Combined pseudocolor image.

α is usually set as a bit of value, such as 0.01, to extract a few and significant change pixels.

As shown in Fig. 1, after extracting large-scale intensity change, the Berger estimator [16], [19] in (5) is used to detect more minor scale changes, such as tire imprints on soft soil. The pdf of $\hat{\rho}_a$ derived as

$$p\left(|\hat{\rho}_a|; |\rho|, N\right) = (2N-1)_2F_1\left(N, N+1/2; 1; |\rho|^2|\hat{\rho}_a|^2\right) \cdot \left(1-|\rho|^2\right)^N |\hat{\rho}_a| \left(1-|\hat{\rho}_a|^2\right)^{N-3/2} \tag{9}$$

where $\hat{\rho}_a$ is parameterized by the true underlying coefficient ρ and the N samples in the local window. It is noticeable that the denominator in (5) contains a sum rather than a product of two variables in (4); thus, the estimator $\hat{\rho}_a$ suggests a more stable result than $\hat{\rho}_c$. The joint pdf of $\hat{\rho}_a$ and $\hat{R}$ can be referenced at [19]. With the above steps, the output of the noncoherent intensity change detection and the complex coefficient estimation is combined into the final difference image, which outstands the large-scale and small-scale changes.

After getting the difference image, we try to concatenate the master image f, the slave image g, and the difference image along the channel dimension. The combining process of one patch is shown in Fig. 2. There are two cars in the smooth region in the master image in Fig. 2(a). These cars moved randomly between the acquiring interval and were parked in their original position in the slave image of Fig. 2(b). The difference image generated by the two-stage method is shown in Fig. 2(c). The subtle position shift of cars leads to low coefficient, and the vehicle trace caused by car moving also appears in the difference image by our change detection method. The pseudocolor image in Fig. 2(d) introduced more landmark information like car intensity and vegetation

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distribution. In the next section, we use the combined pseudocolor image to realize automatic vehicle trace extraction.

It is noticeable that [35] also tries to construct the pseudocolor image to make use of landmark information. However, in this method, the unsupervised false elimination depends on the landmark category and may not have well robustness to the various scenes. Compared with this method, the pseudocolor in this article does not need to pixelwise classification of the imaging scene and can be operated more efficiently.

B. Vehicle Trace Detection Network Based on Spatial CNN and Attention Mechanism

In the trace detection task, we should classify each pixel in the input image as vehicle trace or background class, e.g., if the size of input pseudocolor image is $512 \times 512 \times 3$, the size of the corresponding detection result is 512×512, where 0 means background and 1 means trace vehicle. Due to the flexible scalability and robustness of Unet [38], we take it as the basic structure and retain its encoder and decoder module. Since vehicle trace properties in pseudocolor images are different from optical images, such as the long continuity and sparse distribution, it is not reasonable to directly use Unet to realize vehicle trace detection. Considering the characteristics of pseudocolor image and vehicle trace, we introduce the SCNN and spatial attention mechanism into the Unet structure, and the final detection network is shown in Fig. 3. In the encoder module, the convolution layer, batch normalization (BN) layer, rectified linear unit (ReLU), and max-pooling layer are hierarchically stacked to extract high-dimensional features from the input image. Given an image patch with the size of $512 \times 512 \times 3$, we will get 512 feature maps with 32×32 units by the encoder module. Then, the SCNN is used to increase the receptive field of the feature maps. After that, the decoder module uses the upsampling layer, convolution layer, and BN layer to retain the original size of the input image and realize the final classification. It is noticeable that the attention mechanism is introduced into the decoder module to improve the feature extraction ability of vehicle trace. The SCNN and attention mechanism in our network is shown in the following.

1) Spatial CNN to Improve the Feature Representability: The traditional CNN usually realizes image information transfer between different layers with layerwise convolution. Although this type of methods has shown strong capability to extract semantics from images with strong appearance coherence, the spatial relationships of pixels across rows and columns of the input images are not fully explored. Thus, the traditional CNNs may not perform well for objects that have long structure region and could be disconnected, such as the vehicle trace region in our CCD pseudocolor images. To address the above issue, the SCNN is introduced into our classification network and it is shown in Fig. 4.

The SCNN can be categorized into four modules as the convolution directions: up to down, down to up, left to right, and right to left [45]. Taking the up to down module as an example, given a feature map of size $C \times H \times W$, where H, W, and C denote the number of rows, columns, and channels,

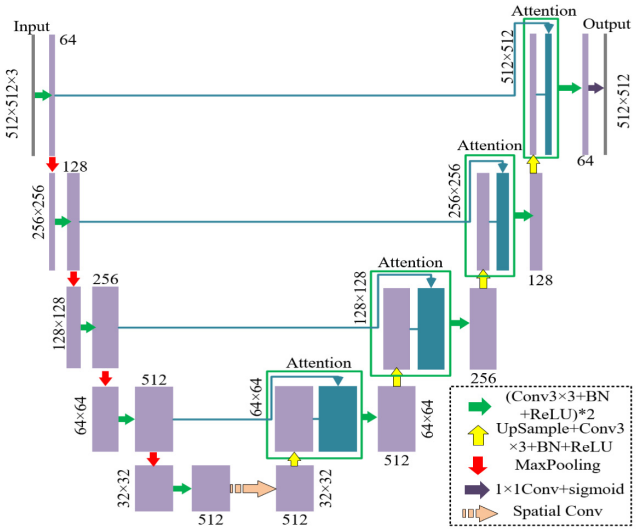


Fig. 3. Structure of the proposed detection network.

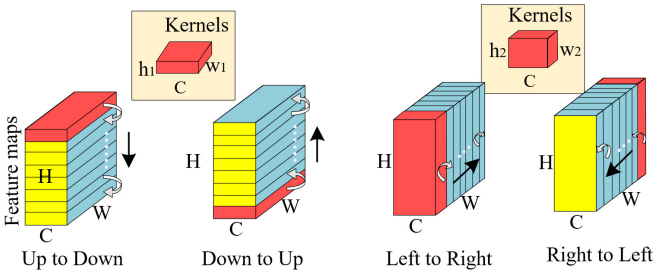


Fig. 4. Structure of the SCNN.

respectively, this feature map is first to split into H slices of size $C \times 1 \times W$. Then, C convolution kernels of size $C \times 1 \times w$ are used to convolve with the first slice of the split feature map. The output of this convolution is added to the next slice, and the generated slice is sent to the next convolution layer until the last slice is updated. It is noticeable that the height of the convolution kernels here is one, and the width is larger than the traditional kernels, such as 9 pixels.

Assuming the feature maps $\mathbf{F}$, the element in $\mathbf{F}$ is denoted as $X_{i,j,k}$, where i, j, and k indicate the indexed channel, row, and column, respectively. Similarly, for the element $K_{i,j,k}$ in the convolution kernel $\mathbf{K}$, i and j are, respectively, the channel i of the last slice and the j channel of the current slice, and k is the offset between two elements, and the convolution process is represented as

$$F'_{i,j,k} = \begin{cases} F_{i,j,k}, & j = 1 \\ F_{i,j,k} + f\left(\sum_m \sum_n F'_{i,j-1,k+n-1} \times K_{m,i,n}\right), & j = 2, 3, \ldots, H \end{cases} \tag{10}$$

where f is the ReLU function. Thus, with the above convolution layer, the semantic information of the top slice in the feature maps flows into the bottom slice of the feature map.

After attaining the feature map through the up to down module, we repeat the SCNN along different directions to increase the receptive field and the feature representability,

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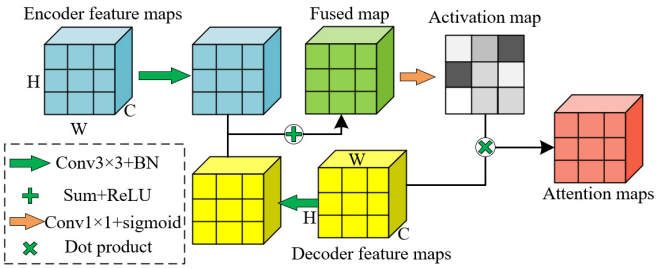


Fig. 5. Structure of our spatial attention module.

as shown in Fig. 4. The output of the former direction is taken as the input of the next direction. It is noticeable that the convolution kernels in one module share the same weight to prevent overfitting. The SCNN is used in the final layer of the encoder module in our detection network to improve the passing efficiency of feature maps, as shown in Fig. 3. Given the input feature maps with $512 \times 32 \times 32$, we can still get the converted feature maps of the same size. Compared with the MRF-based messing-passing methods [46], the SCNN in our network can significantly improve the receptive field along vertical or horizontal direction while keeping a low computation burden.

2) Spatial-Attention Mechanism to Capture Sparse Trace: Unlike the target distribution in optical images, the vehicle trace is usually sparsely distributed at the SAR images, which means that the traditional local-spatial convolution layer pays more attention to the background pixels and cannot fully utilize the valuable trace pixels. Inspired by the success of the attention mechanism for image classification, we introduce the spatial attention [47] into our trace detection network to extract the most informative regions rather than the entire image.

The spatial attention module in our detection network is shown in Fig. 5. Given the encoder feature maps $\mathbf{F}_1$ and decoder feature maps $\mathbf{G}_1$ with size $H \times W \times C$, we first use the convolution layers with 3×3 kernels to convert these feature maps, and then, we use the BN layers to balance the feature distribution. Then, the converted feature maps $\mathbf{F}_2$ and $\mathbf{G}_2$ are summarized over the channels for each spatial position, which indicates the importance of that position, and the fused maps are activated by the ReLU function. After that, one kernel with the size of 1×1 and channel of C convolves with the maps, and the sigmoid function is used to attain the weighting factors for each position. The output from the sigmoid is then used to rescale the magnitude of the activations on the decoder feature maps $\mathbf{G}_1$. Formally, the fused maps $\mathbf{F}_{\text{fuse}}$ are defined as

$$\mathbf{F}_{\text{fuse}} = f((B(\mathbf{F}_1 * \mathbf{K}_1) + B(\mathbf{G}_1 * \mathbf{K}_2)) * \mathbf{K}_3) \tag{11}$$

where $*$ is the convolution operation, B is the BN layer, $\mathbf{K}_1$ and $\mathbf{K}_2$ are the kernels with size 3×3 for encoder and decoder, respectively, and $\mathbf{K}_3$ is the kernels with size 1×1. f is the ReLU function

$$f(x) = \max(0, x). \tag{12}$$

The final output of the spatial attention module is

$$\mathbf{F}_{\text{final}} = \mathbf{G}_1 \otimes \mathbf{F}_{\text{fuse}}. \tag{13}$$

The spatial attention module assigns different importances for different locations with the above steps. For instance, the trace position should have a larger weighting factor while the background position should have a little factor. What is more, since the attention mechanism in our detection network consists of the basic layer in CNN, such as convolution and activation, the entire forward propagation is differentiable, which means that the loss can be backpropagated to the former layers with our attention module.

C. Adaptive Augmentation of Training Images

After constructing the trace detection network, considering the imbalance sample distribution, the focal loss is used to train the model

$$L_{\text{fl}} = \begin{cases} -\alpha(1-y')^{\gamma} \log y', & y = 1 \\ -(1-\alpha)y'^{\gamma}(1-y'), & y = 0 \end{cases} \tag{14}$$

where y is the true value of one sample or pixel, and y' is the predicted value corresponding to y. Compared with the cross-entropy loss, the exponential factor γ makes this loss more sensitive to the hard sample and the imbalance factor α can suppress the imbalance sample distribution, such as the significant quantity difference of trace pixels and background pixels.

To realize the vehicle detection of SAR CCD images based on deep networks, a vast number of labeled samples are required during the training process, which conflicts with the fact that the trace vehicle samples are hard to acquire due to the expensive flight experiments strict operation condition. Data augmentation is a feasible means to overcome this problem. However, the commonly used augmentation strategies, such as blurring, scaling, brightness, or contrast adjusting, may hurt the structure and continuity of vehicle trace. Meanwhile, the registration accuracy [43] and coherence estimator [12] affect the generation of difference image and pseudocolor images. Therefore, we introduce the registration error and estimation windows into the difference image generation steps.

1) Introducing the Registration Errors: The estimated coherence between image pairs can be considered as the product of different decorrelating factors [20]

$$\rho = \rho_{\text{system}}\rho_{\text{processor}}\rho_{\text{baseline}}\rho_{\text{temporal}}\rho_{\text{registration}} \tag{15}$$

where the radar system coherence ρ_{system} and the processing or imaging coherence $\rho_{\text{processor}}$ will contribute very little to the overall decorrelation for a well-devised radar system. The baseline decorrelation ρ_{baseline} can also be ignored for the airborne system while keeping strict repeat-pass imaging geometry [12]. Among these decorrelation factors, the factor $\rho_{\text{registration}}$ significantly affects the entire coherence, and it follows [43]:

$$\rho_{\text{registration}} = \begin{cases} \dfrac{\sin(\pi\mu)}{\pi\mu}, & 0 \le \mu \le 1 \\ 0, & \mu > 1 \end{cases} \tag{16}$$

where μ is the position shift value along azimuth or range direction. According to (16), the registration accuracy better

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than 0.2 pixels in azimuth and range will reduce the coherence by less than 10%.

Although the optical flow-based method [44] used in this article could realize the accurate registration of SAR image pairs within 0.1 pixel of most image samples, the various landmark information and background clutter will affect the registration accuracy, further leading to unsatisfactory decorrelation. To improve the detection robustness, especially for the trace samples in various background regions, we try to introduce the registration errors into the training datasets and train a detection model with high tolerance.

To be specific, for the paired images I_1 and I_2 with coarse registration, the coregistration error is about 1–2 pixels. The fine registration wants to find the displacements between both images, which can be defined by $\mathbf{u} : \mathbf{x} \rightarrow \mathbf{u}(\mathbf{x}) \in \mathbb{R}^2$. It is usually assuming that the pixel intensities remain constant from image I_1 to image I_2, which can be written as [44]

$$I_1(\mathbf{x}) = I_2(\mathbf{x} + \mathbf{u}(\mathbf{x})). \tag{17}$$

For optical flow computation methods, a change in intensity can be linearized as

$$I_2(\mathbf{x} + \mathbf{u}(\mathbf{x})) = I_2(\mathbf{x}) + \nabla I_2 \cdot \mathbf{u}. \tag{18}$$

The used algorithm tries to minimize the criterion

$$J(\mathbf{u}; \mathbf{x}) = \sum \omega(\mathbf{x}' - \mathbf{x}) \left(f_1(I_1(\mathbf{x}')) - f_2(I_2(\mathbf{x}' + \mathbf{u}(\mathbf{x}))) \right)^2 \tag{19}$$

where ω is the local window. f_1 and f_2 are the space projection function for I_1 and I_2, respectively. ∇ means the 2-D gradient of the input image. Using the iterative Gauss–Newton strategy to minimize $J(\mathbf{u}; \mathbf{x})$, we will get the offset o_k between the nearest original point k and the interpolated point, where o_k is the fractional number usually following into $[-1,1]$. Introducing the manual co-registration error o_k^e into o_k

$$o_k' = o_k + o_k^e \tag{20}$$

where o_k^e is the randomly generated error value following the uniform distribution $U(-\tau, \tau)$. τ means the error boundary, which is a little positive value, such as 0.1 and 0.2, and the smaller the absolute value of o_k^e, the higher the coherence of the sample k between paired images. For each offset in the transformation matrix, by adding the generated offset to the original offset, we will attain the offset with various offset errors and further can attain the difference image with different registration errors. By introducing different τ values into the difference image generation of training samples, we can get the coregistration SAR image pairs with various registration errors. The experimental results in Section IV-D demonstrate that using the established samples to train our detection model has good robustness to different registration accuracies and backgrounds.

2) Multiple-Scale Windows for Augmenting Difference Images: Besides the co-registration error, the scale of the estimator window can also affect the estimated coherence in the difference image and pseudocolor image. The two-stage method in Fig. 1 is used to detect the decorrelation caused by trace pixels. Especially, the pdf of estimator $\hat{\rho}_a$ in (9) is shown in Fig. 6. One can see that for underlying coherence $|\rho|$, the estimator $\hat{\rho}_a$ shows different distributions of different estimation windows. The larger the estimator window, the more accurate the coherence estimation. It is noticeable that too large window also leads to unclear trace boundary, which will affect the final detection accuracy [17], [19]. Meanwhile, even with same window size, the estimator $\hat{\rho}_a$ still shows different distribution curves for different underlying coherences $|\rho|$, which means that for the area with different decorrelation coefficients, the same estimator will show different estimation accuracies.

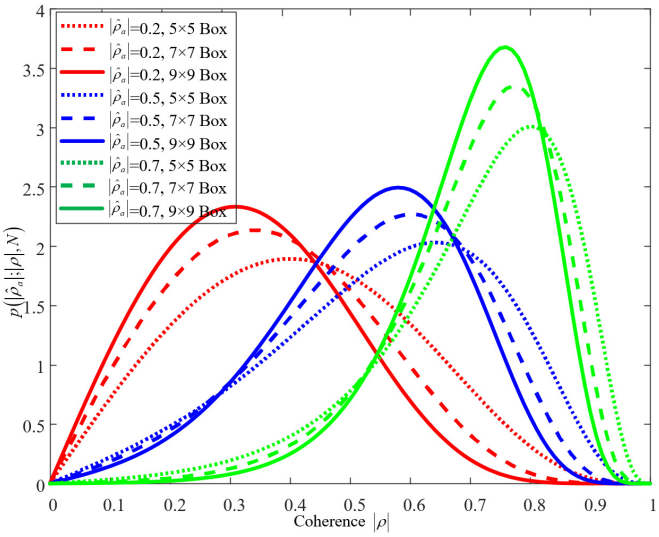


Fig. 6. PDF of estimator $\hat{\rho}_a$ with various window scales.

With the above analysis, we try to use various window scales, like (5,7,9), to estimate the coherence of training image pairs to suppress the estimation difference of different background regions. Replacing the single window scale with multiple scales, the training images are more augmented. It is noticeable that we still use the 5 × 5 window to estimate the coherence of testing image pairs in the testing process.

IV. Experimental Results and Discussion

In our proposed framework, the pseudocolor image is synthesized to provide adequate landmark information, the SCNN, and attention mechanism that are introduced into Unet structure to capture the unique characteristics of vehicle trace, and the manual registration error and estimation strategy based on various windows size are proposed to realize robust vehicle trace detection under different backgrounds. To verify the effectiveness of the proposed method, these three parts are analyzed separately to illustrate each module's contributions to the whole framework.

A. Data Description and Evaluation Criteria

The CCD data used in this article consist of a Sandia SAR dataset and our measured SAR datasets [34]. The Sandia SAR data sample (SSDS) contained 64 SAR complex images with a size of 3000 × 1754 pixels collected by a Sandia National Laboratory test platform in the spotlight mode with

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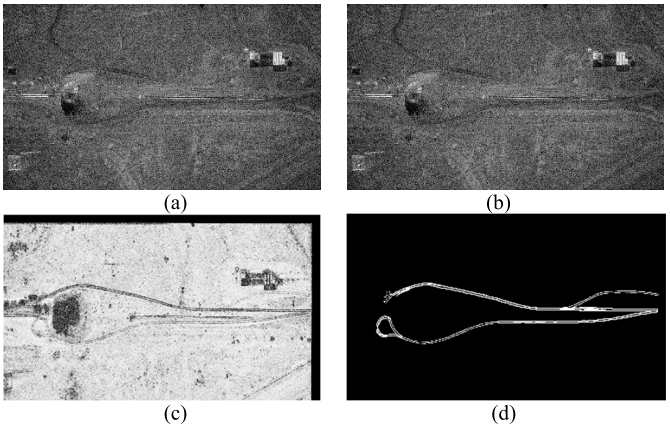


Fig. 7. Image pair from SSDS. (a) Master image. (b) Slave image. (c) Difference image provided by Sandia. (d) Ground truth provided by authors.

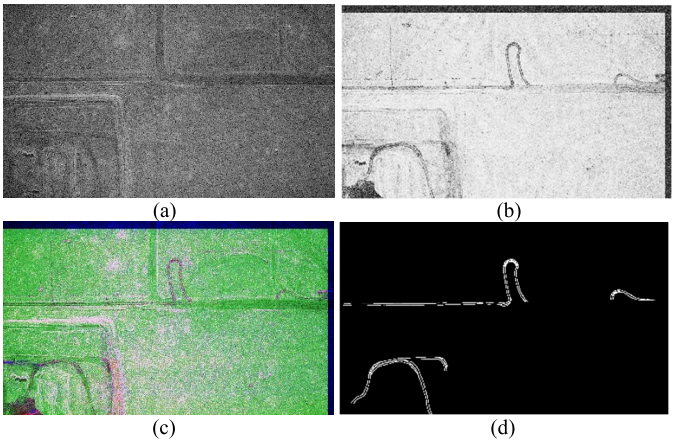


Fig. 8. Image pair from SSDS. (a) Master image. (b) Difference image. (c) Synthesized pseudocolor image. (d) Ground truth provided by authors.

a center frequency of 16.8 GHz at 6-in (0.1524 m) range and cross-range resolution. The collection consists of two flights called simply "am" and "pm." Each of these flights contains 32 spotlight SAR images collected near Albuquerque, New Mexico, and each flight is further broken down into eight flight passes that contain four images per pass. Thus, we can get 32 image pairs from SSDS, and the time interval of each interval is a few hours.

An image pair from SSDS is shown in Fig. 7. Fig. 7(a) shows the master image, while Fig. 7(b) shows the slave image corresponding to Fig. 7(a). The difference image between registered Fig. 7(a) and (b) is generated by Sandia and is shown in Fig. 7(c). One can see that there is continued and parallel vehicle trace in Fig. 7(c); at the same time, there are still other decorrelation pixels in Fig. 7(c) caused by shadow and vegetation. Since SSDS does not provide trace labels of each pair, we manually annotate the vehicle trace pixels according to the difference image and our trace detection task, and the ground truth corresponding to Fig. 7(c) is shown in Fig. 7(d). It is noticeable that the position shift between the image pair in SSDS can be used to validate the effectiveness of our proposed co-registration strategy.

In this article, we choose the former 20 image pairs as the training dataset while the latter 12 image pairs as the testing dataset. Meanwhile, three measured SAR image pairs were collected in this article as the testing dataset to validate the robustness of the proposed method. The detailed instructions of our measured data will be shown as follows.

As the definitions in [35], we define the number of correctly detected trace pixels as true positive (TP), the number of correctly background pixels as true negative (TN), the number of undetected trace pixels as false negative (FN), and the number of wrongly detected background pixels as false positive (FP), and a few evaluation criteria are listed as follows:

$$\text{Precesion} = \text{TP}/(\text{TP} + \text{FP}) \tag{21}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{22}$$

$$\text{F1} = 2/(1/\text{Precesion} + 1/\text{Recall}) \tag{23}$$

where Precesion is the detection accuracy of vehicle trace, shorten for P. Recall is the percentage of correctly detection vehicle trace to the total trace pixels, shorten for R. F1 is an evaluation coefficient that considers both precision and recall. P, R, and F1 fall into [0,1], while the higher the value, the better the detection performance. Meanwhile, we also provide the average running speed of all image pairs based on the platform of i9-11950H CPU and NVIDIA RTX A5000 GPU.

B. Experimental Results of Pseudocolor Image Synthesizing

In Section III-A, we use the two-step method to estimate the coherence between image pairs and the channel concatenation strategy to construct the pseudocolor image. To intuitively understand the effect of the image synthesizing, we choose one image pair from SSDS, and the coherence estimation result and pseudocolor of them are shown in Fig. 8.

From the difference image in Fig. 8, we can see that the coherence estimation method used in this article can realize fine co-registration of the repeat-pass SAR images and possess a good estimation performance. Meanwhile, the vehicle trace in the difference image is prominent and the background region is suppressed. The pseudocolor image in Fig. 8(c) shows that our synthesizing method absorbs the landmark information in repeat-pass images, and the coherence estimation results in the difference image. Thus, the proposed pseudocolor image synthesizing method has well robustness of the SSDS dataset. In the next section, the experimental results of our method, which takes the pseudocolor image in Fig. 8(c) as the input image and the ground truth in Fig. 8(d) as the optimization target, are analyzed in detail.

C. Experimental Results of Trace Detection Network

The effectiveness of the proposed trace detection network based on SCNN and attention mechanism is verified here.

1) Parameter Setting and Feature Visualization: For the training process, we first crop the synthesized pseudocolor image from the training dataset of SSDS into 512×512 pixel patches without overlapping. We can get 1160 training slices and corresponding ground truths using common augmentation strategies, including flipping and rotation. The root mean square (rms) prop strategy optimizes the focal loss in (14). The batch size is 8, and the epoch number is 100. The

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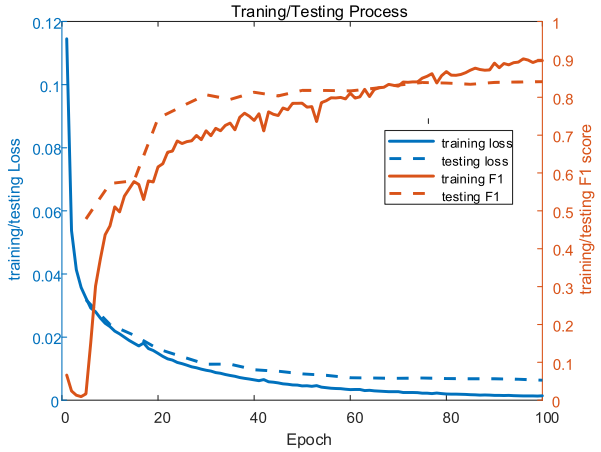


Fig. 9. Training and testing loss of the proposed method.

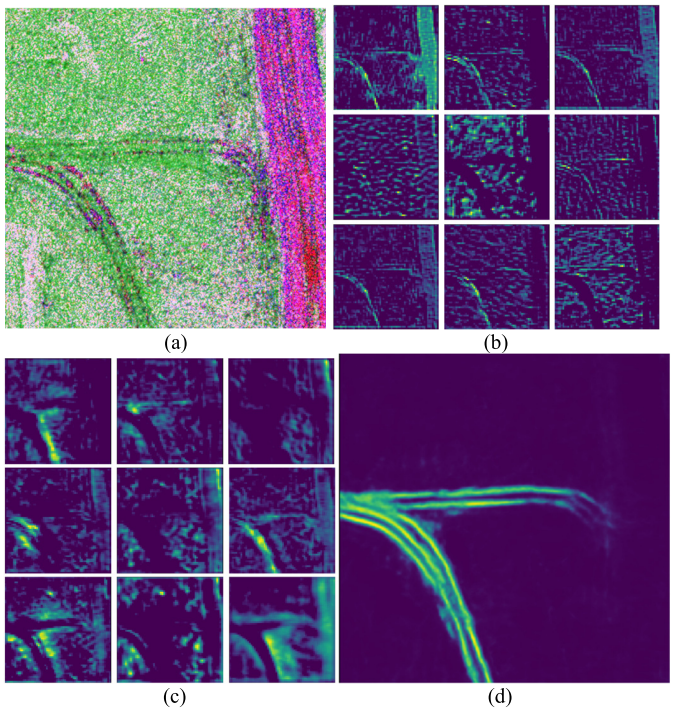


Fig. 10. Input image and corresponding feature maps. (a) Image patch from SSDS. (b) Feature maps of the fourth encoder module. (c) Feature maps of the second decoder module. (d) Heatmap.

initialization learning rate is 0.00001, and the momentum is 0.9 for RMSProp. The weights of our trace detection network are randomly initialized. It is noticeable that we first crop the original pseudocolor image into patches with size 512 × 512 pixels in the testing process. Then, we input these testing patches into the well-trained detection network and combine the patch-based result into the result of the original big image as the cropping order. The training and testing loss of the proposed method is shown in Fig. 9. One can see that the proposed detection network has a stable convergence process. With the increasing epoch, both the training and testing losses achieve the optimum local value, while the training and testing F1 score achieve the highest value. Significantly, the highest testing F1 score of the proposed method is 84.1196%.

One image patch from SSDS with size 512 × 512 pixels and the corresponding feature maps extracted by our well-trained model are shown in Fig. 10. One can see that the former encoder layers usually extract the appearance structure and contour features, while the latter decoder layers usually extract the abstract and semantic features. The final heatmap generated by the final layer in our detection network in Fig. 10(d) where the bright areas represent the trace region and the dark areas represent the background region demonstrates that the proposed method can extract trace regions from the input image.

TABLE I

TRACE DETECTION RESULTS BY DIFFERENT DETECTION METHODS (%)

Method	P	R	F1	Params(M)	Speed(s)
Unet	77.51	84.01	80.63	**7.77**	**0.96**
ChUnet	79.48	78.38	80.92	49.15	1.89
AtUnet	77.76	**87.60**	82.39	34.88	1.39
NUnet	76.80	87.20	81.67	9.16	1.41
Proposed	**81.70**	86.69	**84.12**	35.30	1.46

2) Comparisons of Different Detection Methods: To validate the effectiveness of the proposed method, a few CNN-based detection methods are chosen as the comparison methods. It is noticeable that there are some sliding window-based CNN methods that try to crop small patches for each pixel in the input image and classify this pixel as trace region or background region. However, these small patch-based methods face a huge computation burden and may not be applied to the practical trace detection task; thus, we only use the Unet-based methods as comparison methods in this article. To be specific, the primal Unet [38], attention Unet (AtUnet) [48], channel Unet (ChUnet) [49], and nested UNet (NUnet) [40] are taken as the comparison methods. These deep learning methods share the same training set as the proposed method. It is noticeable that all these methods take the generated pseudo-color image in Section III-A as an input. These methods are executed based on the Pytorch framework, while the training parameters and network settings of them are consistent with the original program. The quantitative evaluation results of these methods are shown in Table I.

In Table I, we can see that the primal Unet achieves the precision of 77.51%, the recall rate of 84.01%, and the F1 score of 80.63%. Among these comparison methods, the AtUnet attains the highest recall rate of 87.60% and F1 score of 82.39%, which demonstrate that introducing the attention mechanism into the Unet structure improves the detection performance. Especially, the AtUnet has a higher precision and recall rate than ChUnet, which is consistent with the analysis that we should use the spatial-based attention mechanism to capture the sparse distribution of the vehicle trace. Compared with these methods, the proposed method achieves the highest precision of 81.70% and an F1 score of 84.12%. These quantization results prove that the proposed detection network has the best detection performance than other detection networks. These improvements are mainly caused by the SCNN module and the spatial attention mechanism in our detection network. Meanwhile, we also analyze the parameter numbers of these methods. Introducing different network structures or attention mechanisms into the primal Unet, the parameter numbers increase with different degrees. The NUnet has the lowest parameters, while the proposed

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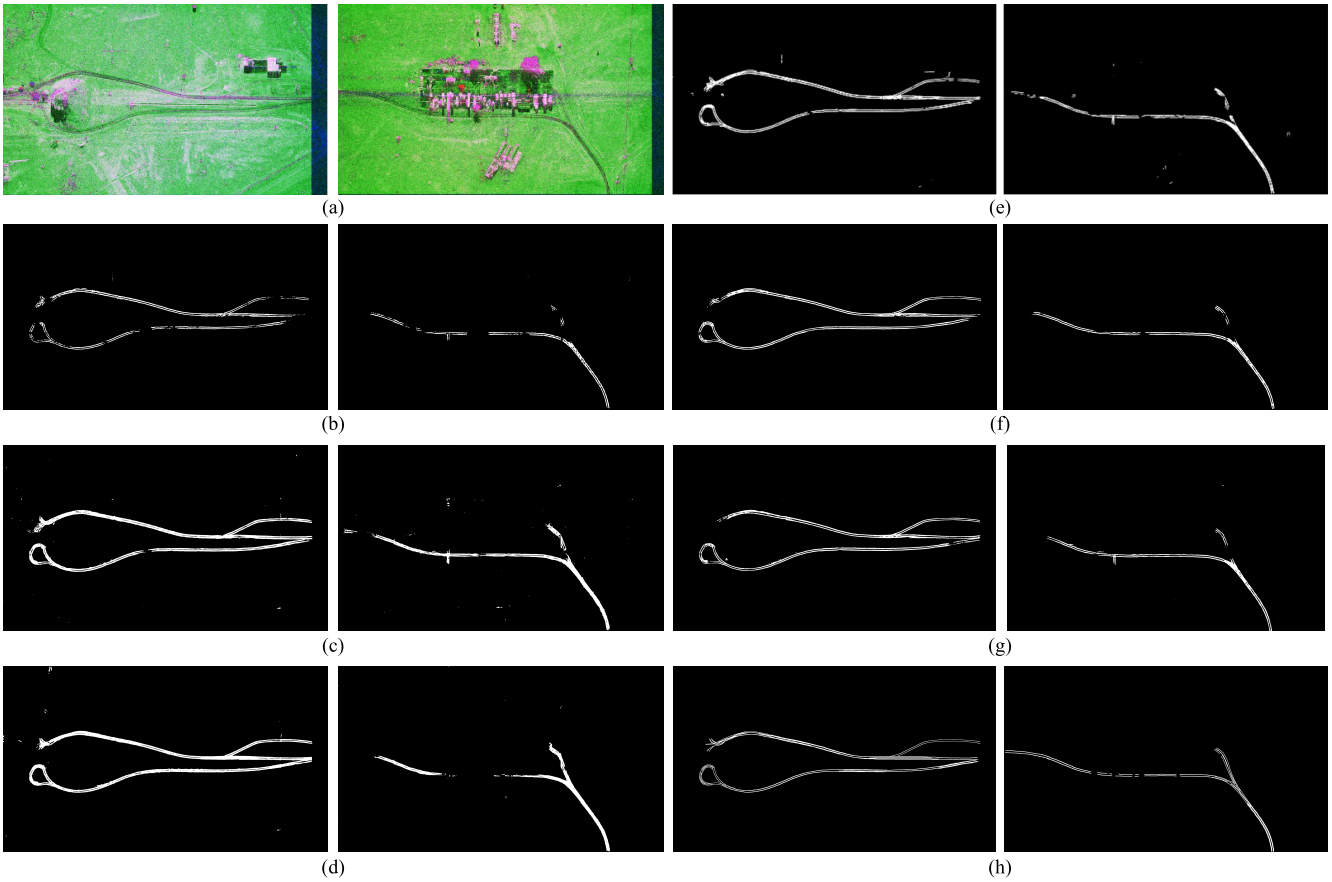


Fig. 11. Input image and corresponding detection results. (a) Two pseudocolor images from SSDS. (b) Unet detection results. (c) ChUnet detection results. (d) AtUnet detection results. (e) NUnet detection results. (f) Proposed method detection results. (g) Proposed method with adaptive augmentation. (h) Ground truths.

method increases the parameter of Unet from 7.77 to 35.50 M. Thus, the increase of weighting parameters makes our model have better characterization ability. We also provide the cost time of the whole SSDS image detection in Table I. The primal Unet has the fastest running speed, and the proposed method can also realize trace detection within 1.5 s for the SSDS image.

The visualization results of the proposed method and other comparison methods are presented in Fig. 11. Two CCD image pairs are selected from SSDS. The generated pseudocolor images and corresponding annotated ground truths are shown in Fig. 11(a) and (h), respectively. From Fig. 11(a), we can see that there is a long continuous vehicle trace in the complicated background, such as buildings and vegetation. The detection results in Fig. 11(b) by Unet extract most trace pixels from pseudocolor images. However, some discontinuous and blurred trace pixels are ignored mainly due to the short operation range of Unet. Compared with Unet, the results in Fig. 11(c)–(e) show that the ChUnet, AtUnet, and NUnet can extract more pixels than the Unet method. Meanwhile, the discontinuous trace pixels can also be extracted by these improved methods. However, the trace boundary of ChUnet and NUnet is not as clear as the result in Unet and AtUnet. Among these methods, our proposed method does not only retain clearer trace contour than other methods but also has the least false alarm in the detection result of Fig. 11(f). Moreover, the quantitative analysis in Table I also declares this point. The above analysis fully demonstrates the effectiveness of the proposed method on the SSDS testing dataset.

TABLE II
DETECTION RESULTS BY INTRODUCING SCNN AND ATTENTION MECHANISM (%)

Method	P	R	F1	Params(M)
Unet	77.51	84.01	80.63	7.77
Unet + SCNN	80.26	85.64	82.86	7.77
Unet+SCNN+Attention	**81.70**	**86.69**	**84.12**	35.30

3) Ablation Study of Trace Detection Networks: The proposed detection method consists of two key modules: the SCNN and the attention mechanism. The superiority of our method has been validated with the above different comparison methods. To further understand each part of our method, we give a detailed performance comparison from Unet to our method. The results are shown in Table II.

a) Effectiveness of the spatial CNN: From Table II, we can see that introducing the SCNN into Unet does not significantly increase the parameter number for that there are a few convolution kernels in the SCNN module. Meanwhile, all the precision, recall rate, and F1 score have been improved with the SCNN. To intuitively understand the effectiveness of SCNN, we choose two image patches from the SSDS. The feature maps without and within SCNN are shown in Fig. 12.

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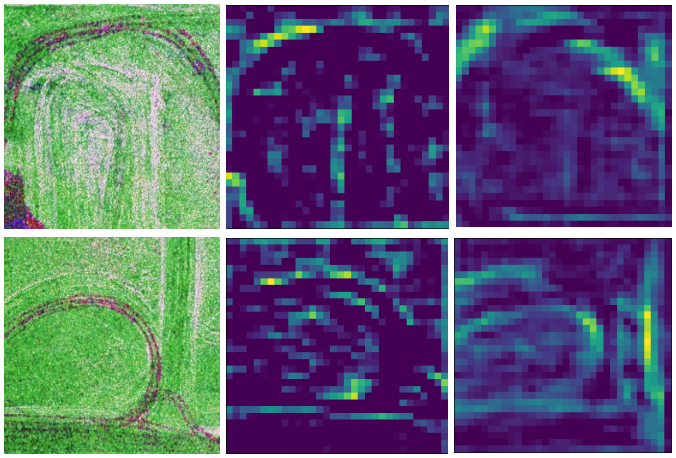


Fig. 12. Input images and feature maps without and within SCNN. (a) Input images. (b) Feature maps before SCNN. (c) feature maps after SCNN.

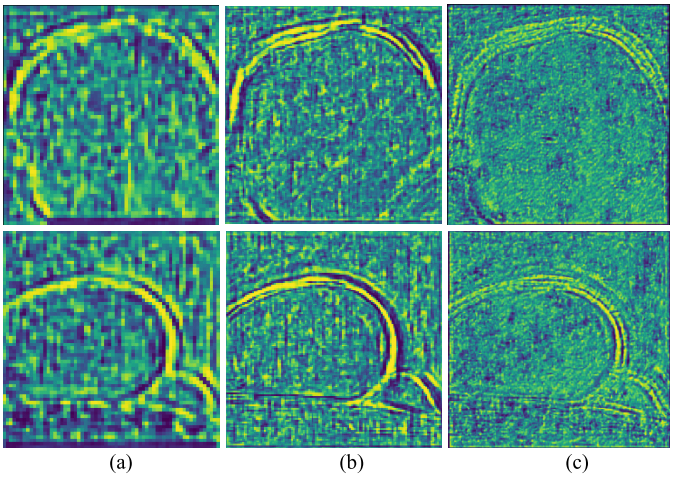


Fig. 13. Attention maps in different attention layers. (a) Former attention maps with size of 64 × 64 pixels. (b) Middle attention maps with size of 128 × 128 pixels. (c) Latter attention maps with size of 256 × 256 pixels.

From Fig. 12, we can see that the feature maps in Fig. 12(b) can well represent the trace pixel distribution in Fig. 12(a). However, these features are too dispersed to show the long continuity of trace pixels. The feature maps within SCNN in Fig. 12(c) show that the converted features have better and smoother continuity. Meanwhile, Fig. 12(c) can better represent the trace pixel distribution in Fig. 12(a). Thus, with the above quantitative analysis and visualization results, the effectiveness of SCNN has been well verified.

b) Effectiveness of the spatial attention mechanism: From Table II, we can see that introducing the spatial attention mechanism into Unet can improve the precision, recall rate, and F1 score at the cost of increasing parameter quantity. To further understand the effect of our attention mechanism, the attention maps of the different attention layers in our detection network are shown in Fig. 13. The attention maps reveal the weighting factors for each position in the corresponding feature maps, while the lighter the attention maps, the higher the weighting factors. One can see that the light regions in Fig. 13 are matched with the trace pixels in Fig. 12(a). Meanwhile, the former maps in Fig. 13(a) reveal weak position information and strong semantic information, while the latter maps in Fig. 13(c) reveal the strong position and weak semantic information. By utilizing these attention maps, the trace pixels are highlighted, while the background region is weakened. With the above analysis, both the quantitation results and visualization results demonstrate the effectiveness of our attention mechanism.

c) Effectiveness of the pseudocolor image: To understand the effect of the pseudocolor image, we replace the input image with the difference images in Section III-A. Meanwhile, the input channels of these detection networks are also reduced from 3 to 1. The results within the difference image are shown in Table III. One can see that the precision, recall rate, and F1 score of these methods have been reduced to a certain extent than the results in Table I. Especially, the F1 score of the ChUnet decreases by nearly 4%, while the F1 score of the proposed detection network also decreases by nearly 3%. These quantitative results show that the pseudocolor image has a better performance than the difference image for the Unet-based detection network. These comparison results further demonstrate the effectiveness of our method.

TABLE III
DETECTION RESULTS BY TAKING DIFFERENCE IMAGE AS AN INPUT (%)

Method	P	R	F1	F1 change rate
Unet	75.73	83.14	79.26	-1.37
ChUnet	76.29	77.82	77.05	-3.87
AtUnet	74.86	86.21	80.13	-2.25
NUnet	75.08	83.56	81.67	-2.58
Proposed	78.31	84.42	81.25	-2.87

D. Experimental Results of Adaptive Augmentation

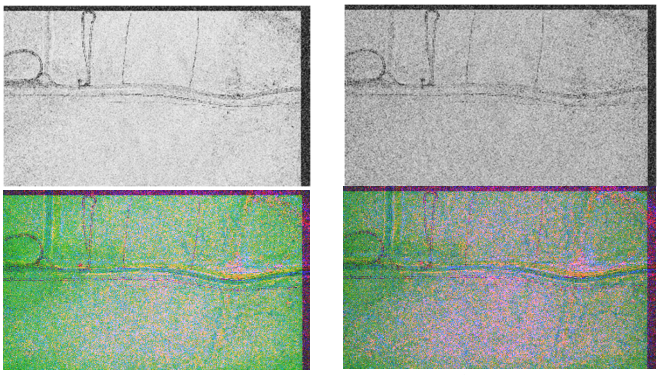
In Section III-C, we proposed the adaption augmentation strategy to increase the number of training samples. In this augmentation, the various registration errors and estimation window scales are introduced into the generation of difference image, while the results based on our detection network are analyzed in this section.

1) Effectiveness of Multiple Registration Errors: For the first augmentation strategy, we assume that all the registration errors for each pixel by our registration method are 0. Then, we add the randomly generated offset to the coordinate transformation matrix. The offsets such as 0.1 and 0.2 are taken as the registration errors. In the augmentation process, we first generate the pseudocolor images with no errors, and then, we generate again the pseudocolor images by introducing the above errors. Thus, the training samples have been expanded twice with our augmentation method. The results under different registration errors are shown in Table IV. One can see that by adding the errors with 0.1 to the training samples, the F1 score has been improved from 84.12% to 86.81%. What is more, the registration error of 0.2 has achieved the highest recall rate of 89.60% and the highest F1 score of 87.67%, which is 3.55% higher than the nonaugmented method. It is noticeable that the F1 score began to decline by increasing the registration error to 0.3 and 0.5. The reason mainly lies in that too high registration error will significantly affect the coherence of vehicle trace samples in the difference

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TABLE IV
DETECTION RESULTS UNDER DIFFERENT REGISTRATION ERRORS (%)

Error boundary	P	R	F1 score	F1 change rate
0	81.70	86.69	84.12	0
0.1	85.07	88.63	86.81	+2.69
0.2	85.82	**89.60**	**87.67**	**+3.55**
0.3	**86.19**	88.50	87.33	+3.21
0.5	84.76	88.24	86.46	+2.34



(a) (b)

Fig. 14. Difference images and pseudocolor images under different registration errors. Error with (a) 0.2 and (b) 0.5.

TABLE V
DETECTION RESULTS UNDER DIFFERENT ESTIMATION WINDOWS (%)

Window Size	P	R	F1 score	F1 change rate
5	81.70	86.69	84.12	0
7	82.91	86.54	84.69	+0.57
9	80.75	87.33	83.91	-0.21
5, 7	**86.41**	**88.17**	**87.28**	**+3.26**
7, 9	85.19	86.78	85.98	+1.76
5, 7, 9	85.44	87.59	86.50	+2.38

images and pseudocolor images, further affecting the final trace detection results, as shown in Fig. 14. Compared with the trace samples in Fig. 14(b) with error 0.5, the trace samples in Fig. 14(a) with error 0.2 are easier to distinguish from the background region.

2) Effectiveness of Multiple Estimation Windows: The results under different estimation windows are shown in Table V. For the single window size, the window with 7×7 achieves the highest F1 score of 84.69% than the window size of 5×5 and 9×9. We also use the combination of 5×5 and 7×7, 7×7, and 9×9 to augment the training samples, respectively. One can see that the combination of 5×5 and 7×7 achieves the highest precision, recall rate, and F1 score of 87.28% than the single scale methods and other combination methods. We also use the combination of 5×5, 7×7, and 9×9 to augment training samples, while the F1 score of 86.50% is lower than that of 87.28%. Thus, the combination of 5×5 and 7×7 window size is appropriate for the training of trace samples, while the 9×9 window size will affect the final detection performance. The reason mainly lies in that the combination of 5×5 and 7×7 effectively considers the accuracy of estimation coherence and the boundary characteristics of trace samples. The above results demonstrate the effectiveness of our multiple window scale methods, and the window scale of 5×5 and 7×7 is a good choice for the learning of vehicle trace detection networks.

TABLE VI
DETECTION RESULTS UNDER DIFFERENT AUGMENTATION STRATEGIES (%)

Method	P	R	F1 score
Without Augmentation	81.70	86.69	84.12
Multiple Errors	85.82	89.60	87.67
Multiple Windows	**86.41**	88.17	87.28
Combing Augmentation	86.25	**89.73**	**87.96**

3) Results Combining Multiple Registration Errors and Estimation Windows: With the above experimental results, utilizing the registration error boundary of 0.2 and the window scale of 5×5 and 7×7 to augment the training samples, the performance on the testing samples can be significantly improved. Thus, we try to use both methods to augment the samples simultaneously and use the augmented images to train our detection network. The results of these augmentation strategies are shown in Table VI. One can see that the combined augmentation method achieves the highest recall rate of 89.73% and F1 score of 87.96%, further demonstrating the effectiveness of our augmentation strategy. The quantitation results of our method with this augmentation strategy are shown in Fig. 11(g). One can see that this method accurately extracts the trace contour and boundary from the input image while keeping a low false alarm.

E. Experimental Results on the Measured SAR Data

The above experimental results on the SSDS demonstrate the effectiveness of our proposed vehicle trace detection method. However, the imaging scene in testing data and training data of SSDS is too ordinary and smooth. To validate the robustness and adaptability of the proposed method under various scenes and radar systems, we select three CCD image pairs that contain various vehicle trace samples and different imaging scenes. The reference images of these pairs are shown in Fig. 15(a). These images are acquired by a Ka-band radar with HH polarization and stripmap mode, and both the range resolution and azimuth resolution are 0.3 m $\times$ 0.3 m. The first and second image pairs covering the sand and vegetation region are acquired at the Baotou city of China, while the third image pair covering snow and the hilly area is acquired at the Zhangjiakou city of China. The scales of these co-registered image pairs are 4778×3436, 7398×5329, and 3516×2538, respectively. All the time intervals between the repeat-pass images of these image pairs are nearly 2 h, while we arranged for a jeep to drive along an irregular route in these regions.

The pseudocolor images in Fig. 15(b) containing vehicle trace pixels and various landmark information demonstrate that our registration method and coherence estimation method can realize well complex coherence coefficient estimation. Meanwhile, the appearance of vehicle trace samples in the pseudocolor image looks very similar to that of SSDS images in Fig. 8. Thus, we can use our well-trained detection model to infer these measured images. To be specific, we use the well-trained model of F1 score 87.96%, which takes our

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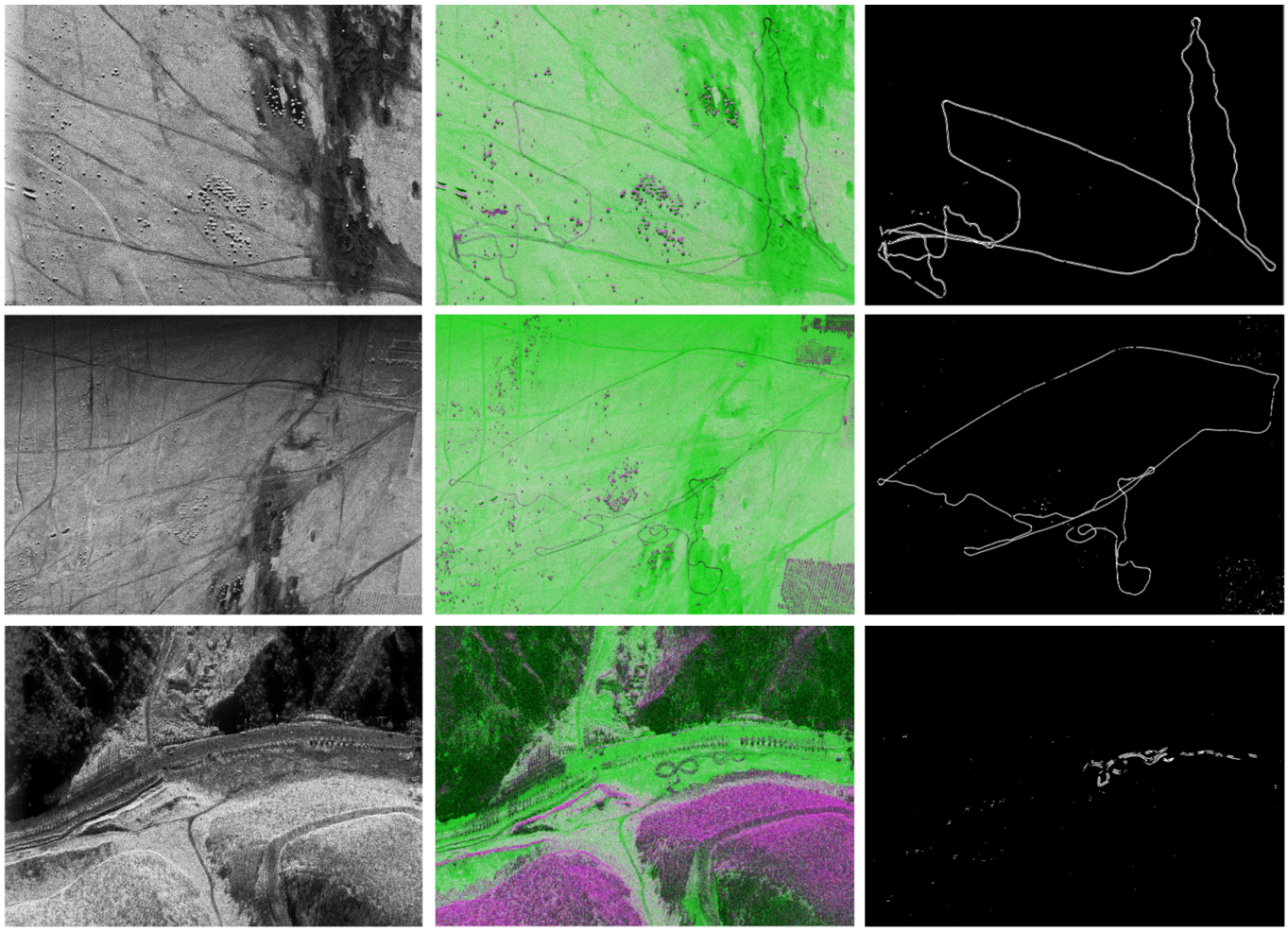


Fig. 15. Detection results on our measured SAR CCD data. (a) Reference image. (b) Pseudocolor image. (c) Detection result.

detection network, including SCNN and attention mechanism as the detection framework, the pseudocolor images in SSDS as an input, and the adaptive augmentation strategy as the auxiliary learning algorithms, to test the measured images. Therefore, we have not retrained the detection network using the measured data. The visualization vehicle detection results in Fig. 15(c) show that our model can accurately extract the long-continuity and parallel trace samples from the pseudocolor images in Fig. 15(b). More specifically, the structure and appearance of vehicle pixels are successively detected in the first and second image pairs, while keeping a low false alarm rate. For the third image pair, although there are some false detected pixels, some trace pixels are also neglected by our method. The reason mainly lies in that the complicated background leads to the appearance and structure degeneration of trace pixels.

The quantitation results corresponding to Fig. 15 are shown in Table VII. During the evaluating process, we first manually label the trace pixels in the difference image and then use the labeled ground truth to compute the precision and recall rate. One can see that for the first and second image pair, the precision is higher than 92% while the recall rate is close to 80%. Meanwhile, the F1 score of these image pairs is close to that of SSDS data, which means that the trace distribution in these measured images is similar to that of SSDS images. Especially, the precision and recall rate of the third image pair is 50.47% and 75.10%, respectively, which are identified in the visualization results in Fig. 15. With the above analysis, although the measured data have different resolution and imaging conditions to the training samples in SSDS, our proposed method can still accurately extract trace samples from input CCD data. It is noticeable that the trace background and landmark could affect the final trace detection performance. We also give the running speed of our method on these measured SAR CCD images in Table VII. One can see that the well-trained detection model has a stable and fast inference speed for these large-scene images.

TABLE VII
DETECTION RESULTS OF OUR MEASURED SAR CCD DATA (%)

Image Pair	P	R	F1	Speed (s)
1	92.85	79.26	85.52	4.26
2	92.36	80.46	86.00	10.04
3	75.10	50.47	60.37	2.13

V. CONCLUSION

To automatically extract vehicle trace samples from the repeat-pass repeat-geometry SAR images, the deep learning-based detection method is detailed and analyzed in this article.

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For the preprocessing of attained SAR image pairs, we first realize the coarse registration of these images within the accuracy of 1–2 pixels. The optical-flow method is used to register the image pairs with subpixel accuracy. Then, we construct the pseudocolor image by combining the repeat-pass images and the difference image. The pseudocolor image does not only cover landmark information but also contains the trace samples in the difference image. Then, considering the long-continuity and parallel distribution of vehicle trace samples, the spatial information enhanced Unet is constructed by introducing the SCNN and the spatial attention mechanism. To improve the robustness to complicated background regions, we present the random registration errors into the registration of image pairs to augment the training samples. Meanwhile, we also realize the coherence estimation with multiple window scales to reduce the model's sensitivity to a complex decorrelation environment. Using the well-trained detection model to infer the testing images, our method achieves the highest detection F1 score than other methods. The ablation study also demonstrates the effectiveness of each module in our method. What is more, three measured SAR CCD image pairs by our imaging system are also analyzed and tested by our detection method in this article. The experimental results show that our method does not only perform good robustness to various imaging scenes but also has good operation efficiency for large-scale images.

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